

large area of base. The choice between the aforementioned methods depends on what you want your robot to do. For a robot, which must keep and pick up things from table, you need to have good height and thus method two is apt, while for a line follower where big height is of no use, the first method is the best.

Before we go on to design the body of the robot, we need to learn some basic aspects of locomotion.

3.2 Types of wheeled locomotion systems

3.2.1 Differential drive

This is perhaps the most commonly used type of locomotion systems in mobile robots, because it is simple and easy to design as well as implement. This type of structure balances itself on three wheels; two of them are motor driven, while the third one is free. The free wheel, also called pivot wheel, can be mounted at the front or at the back of the pair of wheels driven by the motors. Practically, the pivot wheel is best mounted at the back, so it acts as a ‘trailing’ wheel and not as a ‘steering’ wheel.

When both the motor driven wheels rotate in the same direction, the robot moves straight. If both wheels rotate in clockwise direction, and the robot moves forward, then when both wheels rotate in anticlockwise direction, the robot will move backward.

The robot turns when the motor driven wheels move in opposite direction. When left wheel moves clockwise and right wheel moves anticlockwise (as seen from the right side of robot), the robot turns right. On the other hand, when right wheel moves clockwise and left moves anticlockwise (again, as seen from right side of the robot), the robot turns left.

